From Surplus Value to Surplus Dynamics — Mini-Textbook

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1. Orientation

The Problem the Framework Addresses

Traditional economics often tracks a single form of "excess" — the difference between what is produced and what is directly required. That single surplus is then used to explain growth, crises, and instability.

This framework asks a broader question:

- **What happens when *different kinds* of surplus build up at the same time, across multiple domains of the economy and society?**
- **What if these surpluses can convert into one another, sometimes amplifying risks and sometimes dissipating them?**

The problem is that focusing on one surplus alone misses important pathways of instability. A system may appear stable in one dimension (say, material goods) but be on the edge in another (such as finance, rules, or collective attention). When surpluses overflow their "absorption capacity," they create pressures that can spread, trigger feedback loops, and generate crises.

From Single Surplus to Multi-Domain Pressures

The framework generalizes the idea of surplus into **five domains**:

1. **Material surplus** – excess goods or inventories relative to what can be sold.
2. **Financial surplus** – leverage, credit claims, or valuation pressures that outpace collateral.
3. **Institutional surplus** – overload of rules, regulations, or governance structures.
4. **Attention surplus** – more messages, ads, or signals than people can absorb.
5. **Cognitive surplus** – decision backlogs or tasks that outstrip mental or organizational bandwidth.

We call these **surplus pressures**. They are measured against each domain's **absorption capacity** — the system's ability to handle the excess without breakdown.

- **Example:**
    - A warehouse with 120 units of goods but only capacity to ship 100 in time → 20 units are surplus pressure.
    - A bank with $200 in loans backed by $150 of reliable collateral → $50 is financial surplus pressure.

Each domain has its own way of absorbing or dissipating surplus. But if one domain is overloaded, stress may **convert** into another. For instance, material overstock can spill into financial strain (firms cannot repay loans) or institutional strain (emergency rules on storage or tariffs).

The Unification Idea: Excess Pressure vs. Absorption

The paper's central unification is this:

> **Surplus and strain can be treated as "excess pressure" relative to absorption capacity.**

This gives us a common language across domains:

- **Surplus pressure (S):** How much strain builds up in a domain.

- **Absorption capacity (C):** How much the system can handle before stress propagates.
- **Excess ratio:** The relation between S and C determines whether tension is mild, manageable, or critical.

If $S \ll C$, the system easily absorbs pressure.
If $S \approx C$, the system is tense but not yet failing.
If $S > C$, overload begins, raising the chance of triggers and conversions.

This framing lets us write down equations that apply equally well to goods, money, institutions, and information. It also opens the door to **dynamic analysis**: tracking how surpluses grow, shrink, or move between domains over time.

Key Ideas (Orientation)

- The model generalizes from a single surplus to **multiple types of excess pressure**.
- Each type has its own **absorption capacity**.
- Crises occur when **pressures outrun capacities and propagate across domains**.

Common Pitfalls

- Thinking surplus means only "extra goods." In this framework, it also includes financial, institutional, attention, and cognitive overloads.
- Believing stability in one domain guarantees system stability. Cross-domain conversions can still spread collapse.
- Forgetting that **capacity matters**: pressure alone is not enough; it must be evaluated relative to absorption.

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3 Quick Checks

1. **Q:** What are the five surplus domains?
   A: Material, Financial, Institutional, Attention, Cognitive.
2. **Q:** What determines whether pressure is critical?
   A: The balance between surplus pressure S and absorption capacity C.
3. **Q:** Why can't we just track one surplus?
   A: Because stress can convert between domains, meaning risk may build invisibly until it cascades.

2. Core Objects & Notation

This section fixes notation, defines the three core indicators (SSI, CRP, SSE), and works through small numeric examples. All definitions and formulas follow the uploaded paper.

Symbols Box (first use)

- **Units / domains / time**
  - i: unit (e.g., sector, firm, platform).
  - $k \in \{M, F, I, A, C\}$: domain = Material, Financial, Institutional, Attention, Cognitive.
  - t: discrete time (e.g., month/quarter).

- **Core state variables**
  - $S_{i,t}^{(k)} \geq 0$: **surplus pressure** in domain k for unit i at time t (e.g., inventory overhang, leverage strain, rule overload, ad load, decision backlog). Units: "pressure" in domain-appropriate measure.
  - $C_{i,t}^{(k)} \geq 0$: **absorption capacity** in that domain (e.g., sell-through bandwidth, capital buffers, enforcement bandwidth, audience time, automation slack). Same units as S.
  - $\varepsilon > 0$: small constant to avoid division by zero (unitless).
- **Indices**
  - $\mathrm{SSI}_{i,t}^{(k)} \in [0,1]$: **Surplus Saturation Index** for unit i, domain k. Higher = tighter against capacity.
  - $\mathrm{CRP}_t \in [0,1]$: **Collapse Readiness Potential** at time t (system-level). Higher = more ready to tip.
  - $\mathrm{SSE}_t \in [0,1]$: **Semantic (normalized Shannon) Entropy** of topic shares at t. Higher = more fragmented/polarized narratives.
- **Inputs to CRP (examples; measured then standardized)**
  - L_t: leverage metric (e.g., debt/collateral, margin usage).
  - G_t: short-run amplification proxy (e.g., momentum of valuations/throughput).
  - B_t: buffers (e.g., capital/liquidity/inventory cushions).
    These three are mapped into $[0,1]$ via a logistic with weights.

2.1 Surplus Saturation Index (SSI)

$$\boxed{\mathrm{SSI}_{i,t}^{(k)} = \min\left\{1,\ \frac{S_{i,t}^{(k)}}{S_{i,t}^{(k)} + C_{i,t}^{(k)} + \varepsilon}\right\} \in [0,1]}$$

Plain-language interpretation (1–3 lines).
This is "pressure vs. capacity" in one fraction. Near 0 means slack; near 1 means pressure is at or beyond what can be absorbed. ε just prevents divide-by-zero.

Units & interpretation.

Quantity	Units	Interpretation
$S_{i,t}^{(k)}$	domain-specific (e.g., units of goods; $; tasks)	How much stress sits in domain k.
$C_{i,t}^{(k)}$	same as S	How much stress the domain can absorb without spillover.
SSI	unitless in $[0,1]$	Near 1 → high tension; near 0 → comfortable.

Worked numeric examples.

1. **Material (warehouse):** $S = 120$ pallets backlogged; $C = 180$ pallets/day ship capacity; $\varepsilon = 1$.
   $\mathrm{SSI} = \min\{1,\ 120/(120 + 180 + 1)\} = \min\{1, 120/301\} \approx 0.399$.
   Moderate tension: capacity still exceeds pressure.
2. **Financial (bank):** $S = \$50$ risk overhang vs collateral; $C = \$30$ buffer; $\varepsilon = 0.1$.
   $\mathrm{SSI} \approx 50/(50 + 30 + 0.1) = 50/80.1 \approx 0.624$.
   High-ish tension: stress close to capacity.
3. **Attention (platform):** ad load $S = 0.7$ hrs/day; user attention budget $C = 0.6$ hrs/day; $\varepsilon = 0.01$.

SSI $\approx 0.7/(0.7 + 0.6 + 0.01) = 0.7/1.31 \approx 0.534$.
Over-stuffed feed: ads push beyond attention capacity.

2.2 Collapse Readiness Potential (CRP)

$$\boxed{\mathrm{CRP}_t = \sigma\left(w_\ell L_t + w_g G_t - w_b B_t + w_0\right), \quad \sigma(x) = \frac{1}{1+e^{-x}} \in [0,1]}$$

Plain-language interpretation (1–3 lines).
CRP aggregates **leverage** and **self-amplification** (which raise risk) and **buffers** (which lower risk), then squashes the score into $[0, 1]$. Higher CRP = readier to tip.

Concrete examples of L, G, B and standardization.

- L_t (leverage): debt/collateral ratio; share of leveraged positions; duration-mismatch index.
- G_t (amplification): short-window growth of valuations/throughput; order-book impact; new-loan acceleration.
- B_t (buffers): capital adequacy, liquidity coverage, inventory slack, throttling capacity.
  Contextual note. In practice, put each raw series on a comparable scale before applying the logistic, e.g. z-score $(x - \mu)/\sigma$ over a rolling window or map to percentiles, then choose weights (w_ℓ, w_g, w_b) by simple regression or expert judgment. This follows the paper's "Map → Fire → Render" guidance to construct CRP from leverage, short-run gains, and buffers.

Worked numeric examples.

- **Example A (banking week):** Suppose standardized $(L, G, B) = (1.0,\ 0.5,\ 0.8)$ and $(w_\ell, w_g, w_b, w_0) = (1.2,\ 0.8,\ 1.0,\ -0.2)$.
  Linear score $= 1.2(1.0) + 0.8(0.5) - 1.0(0.8) - 0.2 = 1.2 + 0.4 - 0.8 - 0.2 = 0.6$.
  $\mathrm{CRP} = \sigma(0.6) \approx 0.645$.
  Moderately high readiness to tip.
- **Example B (inventory-rich retailer):** $(L, G, B) = (0.2,\ 0.3,\ 1.4)$, weights as above.
  Score $= 1.2(0.2) + 0.8(0.3) - 1.0(1.4) - 0.2 = 0.24 + 0.24 - 1.4 - 0.2 = -1.12$.
  $\mathrm{CRP} = \sigma(-1.12) \approx 0.246$.
  Comfortable: buffers dominate.

2.3 Semantic (Normalized Shannon) Entropy (SSE)

Let $p_{t,1}, \ldots, p_{t,K}$ be topic-cluster shares (sum to 1). Define normalized entropy:

$$\boxed{\mathrm{SSE}_t = \frac{-\sum_{j=1}^{K} p_{t,j} \log p_{t,j}}{\log K} \in [0,1]}$$

Plain-language interpretation (1–3 lines).
SSE measures how fragmented/polarized the discourse is. Near 0 = one dominant narrative; near 1 = many equally sized narratives, making coordination harder.

Effective number of topics.
Let $H_t = -\sum_j p_{t,j} \log p_{t,j}$. Then $N_{\mathrm{eff}} = \exp(H_t)$.
Plain words: this is the "as-if" count of equally important topics producing the same entropy. (Contextual note: the paper uses normalized entropy; N_{eff} is the standard info-theory interpretation of H.)

Worked numeric examples.

- **Example C (polarized):** $K = 4$, shares $p = (0.45, 0.45, 0.05, 0.05)$.
  $H = -(0.45\ln 0.45 + 0.45\ln 0.45 + 0.05\ln 0.05 + 0.05\ln 0.05) \approx 1.12$.
  $\log K = \log 4 \approx 1.386 \Rightarrow \mathrm{SSE} \approx 1.12/1.386 \approx 0.81$.
  $N_{\mathrm{eff}} \approx e^{1.12} \approx 3.06$.

High fragmentation; ~3 effective topics.

- **Example D (coordinated):** $K = 4$, shares $p = (0.85, 0.05, 0.05, 0.05)$.
  $H \approx -(0.85\ln 0.85 + 3 \times 0.05\ln 0.05) \approx 0.58 \Rightarrow \text{SSE} \approx 0.58/1.386 \approx 0.42$.
  $N_{\text{eff}} \approx e^{0.58} \approx 1.79$.
  One big narrative plus minor chatter.

2.4 Putting them together (quick reading)

- **SSI** tells you *local tension* in each domain: pressure vs. capacity.
- **CRP** summarizes *system-wide readiness to tip* from leverage and amplification, net of buffers.
- **SSE** indicates *narrative fragmentation*, which impairs coordination and absorption.

These three indicators are later combined in the "black-hole" early-warning rules.

Key Ideas (Section 2)

- **Same shape, different domains:** Measure a comparable strain signal S against its capacity C via **SSI**.
- **System readiness: CRP** is a logistic blend of leverage (L), short-run gains (G), and buffers (B).
- **Coordination cost: SSE** captures how narrative fragmentation raises friction for collective responses.

Common Pitfalls

- Treating a big S as dangerous without checking C. It's the **ratio**, not the level, that matters (via SSI).
- Building CRP from raw, incomparable series. **Standardize** L, G, B before weighting. (Contextual note aligned with "Map → Fire → Render".)
- Reading SSE as "bad news." It is an **indicator of fragmentation**, not of sentiment per se.

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3 Quick Checks

1. **What does SSI near 1 mean?**
   Pressure is at/over capacity; triggers and spillovers are likely.
2. **How does CRP change if buffers rise, holding L, G fixed?**
   It falls, because buffers enter with a minus sign before the logistic.
3. **When is SSE largest?**
   When topic shares are close to equal, indicating fragmentation and harder coordination.

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3. The Minimal Dynamic Law

This section states the one-period update for surplus pressure, explains each component in plain language, and works through toy numbers in several domains. All definitions follow the uploaded paper.

Symbols Box (first use in this section)

- S_t: stacked **surplus pressure vector** across domains at time t (e.g., $[S_t^{(M)}, S_t^{(F)}, S_t^{(I)}, S_t^{(A)}, S_t^{(C)}]^\top$). Nonnegative. Units: domain-specific.
- $\Gamma(X_t)$: **generation** of new surplus pressure during $t \to t{+}1$; can depend on observables X_t (productivity/boom, valuation shock, traffic surge). Same units as S.
- $A(S_t; C_t)$: **absorption/dissipation** as a function of current pressure S_t and capacities C_t. Nonnegative, componentwise.
- Λ: **conversion matrix** routing pressure across domains; **row-sub-stochastic** (each row sums ≤ 1), so routing cannot create net surplus. Units: fractions.
- $L_t \geq 0$: **losses/write-offs** (e.g., liquidation, default resolution, abandonment). Same units as S.

3.1 Surplus Update Equation

$$\boxed{S_{t+1} = S_t + \Gamma(X_t) - A(S_t; C_t) + \Lambda S_t - L_t}$$

Plain-language interpretation (1–3 lines).
Next period's pressure equals **what you had** S_t, **plus** what was **generated**, **minus** what was **absorbed**, **plus** what was **converted in** from other domains, **minus** what was **lost/written off**. Routing via Λ can move pressure around but cannot create it on net.

3.2 Components and Economic Meaning

(a) Generation $\Gamma(X_t)$

- **What it is.** New stress created this period (e.g., production overshoot, valuation run-up, rule additions, ad push, backlog of decisions).
- **Examples.**
  - **Material:** a surprise output boom adds unsold inventory.
  - **Financial:** a valuation jump increases margin calls risk.
  - **Attention:** marketing campaign increases ad load beyond user time.
- **Shape.** Can be linear or S-shaped in drivers (e.g., accelerates when demand conditions are hot).

(b) Absorption $A(S_t; C_t)$

- **What it is.** How much stress is dissipated using capacities/buffers this period (sell-through, capital buffers, enforcement bandwidth, throttling, automation).
- **Properties.** Increases in S_t raise potential absorption up to the capacity ceiling; higher C_t generally raises absorption. Often modeled with piecewise/threshold or saturating forms.

(c) Conversion ΛS_t

- **What it is.** Routing of stress across domains (e.g., material overhang → financial strain; institutional overload → attention/cognitive drag).
- **Accounting rule.** Λ is **row-sub-stochastic**: rows sum ≤ 1 so **pure conversion cannot increase total** $1^\top S$. This encodes "no perpetual motion" of surplus via routing alone.

(d) Losses L_t

- **What it is.** Irreversible removal of pressure (write-downs, liquidation, policy amnesties, abandoning

tasks, dropping features).

- **Role.** Larger L_t lowers future pressure but can carry real costs (e.g., bankruptcies, churn).

3.3 Toy Numbers (worked, cross-domain)

We illustrate with **two domains** (Material M, Financial F) to keep arithmetic transparent. Suppose:

- $S_t = \begin{bmatrix} S_t^{(M)} \\ S_t^{(F)} \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \end{bmatrix}$.
- **Generation:** $\Gamma = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ (extra production + valuation heat).
- **Absorption:** $A(S_t; C_t) = \begin{bmatrix} a_M \\ a_F \end{bmatrix}$ with
  $a_M = \min\{S_t^{(M)}, C_t^{(M)}\} = \min\{10, 7\} = 7$,
  $a_F = \min\{S_t^{(F)}, C_t^{(F)}\} = \min\{6, 5\} = 5$.*
- **Conversion:** $\Lambda = \begin{bmatrix} 0.10 & 0.05 \\ 0.20 & 0.10 \end{bmatrix}$ (rows sum 0.15 and 0.30 ≤ 1). Then
  $\Lambda S_t = \begin{bmatrix} 0.10 \cdot 10 + 0.05 \cdot 6 \\ 0.20 \cdot 10 + 0.10 \cdot 6 \end{bmatrix} = \begin{bmatrix} 1.3 \\ 2.6 \end{bmatrix}$.
- **Losses:** $L_t = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$.

⇒

$$S_{t+1} = \underbrace{\begin{bmatrix} 10 \\ 6 \end{bmatrix}}_{\text{carry}} + \underbrace{\begin{bmatrix} 4 \\ 3 \end{bmatrix}}_{\Gamma} - \underbrace{\begin{bmatrix} 7 \\ 5 \end{bmatrix}}_{A} + \underbrace{\begin{bmatrix} 1.3 \\ 2.6 \end{bmatrix}}_{\Lambda S_t} - \underbrace{\begin{bmatrix} 1 \\ 0.5 \end{bmatrix}}_{L} = \begin{bmatrix} 7.3 \\ 6.1 \end{bmatrix}.$$

Reading the numbers (plain language).
Material pressure **falls** from 10 → 7.3: absorption (7) and losses (1) more than offset generation (4), partly cushioned by converted-in stress (1.3). Financial pressure **rises slightly** 6 → 6.1: absorption (5) and losses (0.5) almost match generation (3), but conversion from material (2.6) nudges it up. **Routing matters**: even when one domain heals, another can inherit stress.

* *Contextual note.* The min-form is a simple capacity-limited absorber consistent with the paper's emphasis on capacity/buffer ceilings; other saturating forms are possible and used later in S-mechanisms.

3.4 Quick Single-Domain Illustrations

- **Institutional (rules overload):**
  Start $S_t^{(I)} = 5$. New mandates add $\Gamma^{(I)} = 3$. Enforcement absorbs $A^{(I)} = \min\{S^{(I)}, C^{(I)}=6\} = 5$. No conversion in, $L^{(I)} = 0.5$.
  $S_{t+1}^{(I)} = 5 + 3 - 5 + 0 - 0.5 = 2.5$ (cleanup catches up).
- **Attention (ad load):**
  $S_t^{(A)} = 0.9$ hr; $\Gamma^{(A)} = 0.3$ hr; user time $C^{(A)} = 1.0$ hr $\Rightarrow A^{(A)} = \min\{0.9, 1.0\} = 0.9$. A UI change routes 10% of cognitive backlog into attention: $\Lambda_{C \to A} S_t^{(C)} = 0.1 \times 0.8 = 0.08$. Losses $= 0.05$.
  $S_{t+1}^{(A)} = 0.9 + 0.3 - 0.9 + 0.08 - 0.05 = 0.33$ hr (tension relaxes).

These "back-of-the-envelope" updates match the paper's accounting: **generation − absorption + conversion − losses** by domain.

Key Ideas (Section 3)

- The **difference equation** tracks how pressures evolve with only four levers: **generate, absorb, convert, lose**.
- **Row-sub-stochastic** Λ enforces **no net creation** of pressure via routing alone.
- Cross-domain **conversion can shift risk** even when a source domain improves—watch the system, not just a silo.

Common Pitfalls

- Treating conversion as harmless: even with no net creation, **local spikes** can push a domain over its triggers.
- Modeling absorption without capacity limits: the paper's logic is **capacity-aware**; neglecting caps hides kinks.
- Forgetting losses: write-offs are often the **cleanest reset**, though costly—exclude them and pressure lingers.

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3 Quick Checks

1. **Why can't Λ increase total pressure on its own?**
   Because its rows sum ≤ 1; it **routes** rather than creates net surplus.
2. **Name the four components that change S over one step.**
   Generation Γ, absorption A, conversion ΛS, losses L.
3. **If capacities jump up next period, what happens to $A(S; C)$?**
   It typically **rises**, increasing dissipation and lowering next-period pressure.

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4. Four Mechanisms S1–S4

We now formalize the four simple mechanisms the paper uses to generate rich dynamics with undergrad math: **triggers, feedback, mode switches, and conversion**.

Symbols Box (first use in this section)

- $\mathrm{SSI}_{i,t}^{(k)} \in [0, 1]$: Surplus Saturation Index for unit i, domain k, time t. (From §2.)
- $\alpha_{i,k} > 0$: trigger increment for unit i, domain k.
- $\tau_{i,k} \in (0, 1)$: trigger threshold for SSI.
- $A_+(s)$: S-shaped **amplification** function (gain), increasing and saturating.
- $m_t \in \{\text{normal}, \text{overload}\}$: **mode** at time t. Parameters $(\Gamma^{(m)}, A^{(m)}, \Lambda^{(m)}, L^{(m)})$ depend on m_t.
- $\Lambda = \{\lambda_{j\to k}\}$: **conversion matrix**; rows sum ≤ 1 (row-sub-stochastic). $\delta_{j\to k} \in [0, 1]$: conversion loss fraction.

S1 — Trigger (threshold activation)

Purpose (one line). Turn on extra surplus generation when local tension (SSI) crosses a domain-specific

threshold.

Hard trigger (step)

$$\Delta S_{i,t}^{(k)} \mathrel{+}= \alpha_{i,k}\, \mathbf{1}\{\mathrm{SSI}_{i,t}^{(k)} \geq \tau_{i,k}\}.$$

Plain words. If SSI is high enough, add a fixed bump to pressure this period; otherwise add nothing.

Soft trigger (smooth)

Two standard softeners:

1. **Overshoot:** $\Delta S_{i,t}^{(k)} \mathrel{+}= \alpha_{i,k}\, [\mathrm{SSI}_{i,t}^{(k)} - \tau_{i,k}]_+$, where $[x]_+ = \max\{x, 0\}$.

2. **Logistic step:** $\Delta S_{i,t}^{(k)} \mathrel{+}= \alpha_{i,k}\, \sigma\left(\beta(\mathrm{SSI}_{i,t}^{(k)} - \tau_{i,k})\right)$ with slope $\beta > 0$.
   Plain words. The closer SSI is to the threshold (or beyond), the larger the incremental bump, smoothly. (Logistic echoes the paper's use of logistic shapes.)

Numeric example (institutional).
Given $\mathrm{SSI}^{(I)} = 0.64$, threshold $\tau = 0.6$, $\alpha = 1.5$.

- **Hard:** add 1.5 (since $0.64 \geq 0.6$).
- **Overshoot:** add $1.5(0.64 - 0.6) = 0.06$.
- **Logistic (β=20):** $\sigma(20 \cdot 0.04) = \sigma(0.8) \approx 0.69 \Rightarrow$ add $1.5 \times 0.69 \approx 1.04$.
  Interpretation. Hard trigger flips on full blast; overshoot is tiny; logistic is in-between.

S2 — Feedback (S-shaped amplification)

Purpose (one line). Allow gains to rise with pressure but saturate at high levels.

Examples of $A_+(s)$ (increasing, concave at top)

- **Rational form:** $A_+(s) = \dfrac{r\, s}{1 + \alpha s}$ with $r, \alpha > 0$.

- **Logistic form:** $A_+(s) = \dfrac{K}{1 + e^{-(as-b)}} - \dfrac{K}{1 + e^{b}}$ (shifted so $A_+(0) = 0$).
  Plain words. Small pressure yields little gain; mid-range ramps up; high pressure saturates. This captures self-reinforcing booms that cannot grow without bound.

Numeric example (financial).
Let $s = S_t^{(F)} = 6$, $r = 0.9$, $\alpha = 0.2$.

$$A_+(6) = \frac{0.9 \cdot 6}{1 + 0.2 \cdot 6} = \frac{5.4}{2.2} \approx 2.455.$$

Interpretation. With moderate financial heat, amplification adds about 2.46 units before saturation tempers it.

S3 — Role/Structure Shift (mode switching)

Purpose (one line). Switch parameter sets when the system is overloaded—e.g., rules, bargaining power, or operating protocols change.

Mode variable and piecewise parameters

Let $m_t \in \{\text{normal}, \text{overload}\}$. Use thresholds (often on SSI or a composite) to set

$$(\Gamma, A, \Lambda, L) = \begin{cases} (\Gamma^{(\text{normal})}, A^{(\text{normal})}, \Lambda^{(\text{normal})}, L^{(\text{normal})}), & \text{if } \max_k \text{SSI}_t^{(k)} < \tau_m, \\ (\Gamma^{(\text{overload})}, A^{(\text{overload})}, \Lambda^{(\text{overload})}, L^{(\text{overload})}), & \text{otherwise.} \end{cases}$$

Plain words. Crossing a stress threshold flips the "operating rulebook": capacities, buffers, routing, or write-off policies shift.

Concrete sketches (aligned with paper's domains).

- **Banking:** in overload, $\Gamma^{(F)}$ falls (new lending throttled), $A^{(F)}$ kinks (buffers cap out), $L^{(F)}$ rises (resolution), and Λ channels more real→financial stress.
- **Warehousing:** overload tightens $A^{(M)}$ (dock throughput caps), boosts $L^{(M)}$ (discounting/clearance), and may route $M \to F$ (financing strain).
- **Platforms:** overload raises attention/cognitive S, prompting a **policy shift** (reduce ad load, increase throttling), i.e., $\Gamma^{(A)} \downarrow$, $A^{(A)} \uparrow$.

Numeric example (banking mode switch).
Threshold $\tau_m = 0.6$, $\max \text{SSI}_t = 0.68 \Rightarrow m_{t+1} = \text{overload}$.
Suppose in **normal**: $\Gamma^{(F)} = 3.0,\ A^{(F)} = 5.0,\ L^{(F)} = 0.5$.
In **overload**: $\Gamma^{(F)} = 1.0,\ A^{(F)} = 3.5$ (capacity kink), $L^{(F)} = 1.5$ (accelerated resolution).
If $S_t^{(F)} = 6$ and $\Lambda S_t^{(F)} = 2.6$ (unchanged), then

- **Normal:** $S_{t+1}^{(F)} = 6 + 3 - 5 + 2.6 - 0.5 = 6.1$.
- **Overload:** $S_{t+1}^{(F)} = 6 + 1 - 3.5 + 2.6 - 1.5 = 4.6$.
  Interpretation. Mode switch trades throughput for stabilization: pressure drops faster via higher losses and lower generation.

S4 — Conversion (routing with losses)

Purpose (one line). Route stress between domains, possibly leaking part of it as true dissipation.

Routing rule with losses

For each destination k,

$$S^{(k)} \leftarrow S^{(k)} + \sum_j \lambda_{j\to k}\,(1 - \delta_{j\to k})\,S^{(j)}, \qquad \text{with } \sum_k \lambda_{j\to k} \le 1.$$

Plain words. A fraction $\lambda_{j\to k}$ of domain j's stress is routed to k; a portion $\delta_{j\to k}$ evaporates (true loss). Row sums ≤ 1 ensure routing cannot create net pressure.

Numeric example (material → financial with leakage).
Let $S^{(M)} = 10$, $S^{(F)} = 6$. Route $\lambda_{M\to F} = 0.20$ with loss $\delta_{M\to F} = 0.25$.
Then added to F: $0.20(1 - 0.25) \times 10 = 0.15 \times 10 = 1.5$.
If $\sum_k \lambda_{M\to k} = 0.9$, the remaining 0.1 is implicitly lost (no destination).
Interpretation. Only part of material stress becomes financial stress; leakage and row-sum ≤ 1 keep totals from exploding.

How S1–S4 combine (intuitive picture)

- **S1** can **ignite** extra pressure when local tension is high.
- **S2** can **amplify** active pressures but saturates.

- **S3** can **flip** operating rules, changing all parameters at once.
- **S4** can **move** pressure around, creating **local hot spots** without raising the global total.
  This quartet is enough to produce multiple attractors and hysteresis in later sections.

Key Ideas (Section 4)

- Thresholds and S-shapes are sufficient to model nonlinearities without advanced math.
- Mode switches formalize "policy/role changes" under stress via piecewise parameters.
- Conversion respects conservation (row sums ≤ 1) and allows leakage via δ.

Common Pitfalls

- Using only linear gains: you'll miss mid-range surges and high-range saturation captured by $A_+(s)$.
- Setting conversion rows to sum > 1: that would **create** pressure, violating the paper's accounting.
- Forgetting that mode switching can change **all** of Γ, A, Λ, L, not just one.

---

3 Quick Checks

1. **What's the difference between a hard and soft S1 trigger?**
   Hard uses a step $\mathbf{1}\{\text{SSI} \geq \tau\}$; soft uses overshoot $[\,\cdot\,]_+$ or a logistic so effects grow smoothly near τ.
2. **Why must $\sum_k \lambda_{j\to k} \leq 1$?**
   So routing doesn't create net pressure; any shortfall is interpreted as leakage/loss.
3. **What changes when m_t flips to overload?**
   Parameters (Γ, A, Λ, L) switch to their overload values—often lower Γ, kinked A, higher L, and altered Λ.

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5. One-Dimensional Reduction & Multiple Attractors

This section shows how the multi-domain system can be "collapsed" into a single order parameter, then demonstrates why **multiple long-run regimes (attractors)** naturally appear.

5.1 Order Parameter (Scalar Reduction)

Let S_t be the vector of all domain pressures at time t. Define a scalar **order parameter** as a weighted sum:

$$\boxed{s_t = w^\top S_t = \sum_k w_k S_t^{(k)} \geq 0}$$

- $w = (w_k)$ are nonnegative weights reflecting domain importance.
- **Plain words.** Instead of tracking five separate $S^{(k)}$, compress them into one scalar index s_t.

⚠️ **Disambiguation of "collapse":** Here "collapse to a scalar" means **dimensionality reduction** (compression into s_t), **not** system failure.

5.2 Toy Map (reduced dynamics)

The paper proposes a simple one-dimensional recurrence:

$$s_{t+1} = s_t + A_+(s_t) - A_-(s_t) + \xi_t,$$

with

$$A_+(s) = \frac{0.9\,s}{1 + 0.2s},$$
$$A_-(s) = \begin{cases} 0.45s, & s \le 6, \\ 0.15s + 1.5, & s > 6. \end{cases}$$

- $A_+(s)$: S-shaped amplification (feedback that rises then saturates).
- $A_-(s)$: piecewise absorption (linear slope at low s; kinked to weaker slope plus fixed loss at higher s).
- ξ_t: small noise/shock.

Plain words. The map says: next period's scalar pressure = current pressure + self-amplification − absorption + random bump.

5.3 Fixed Points (steady states)

A **fixed point** satisfies $s^* = f(s^*)$, where

$$f(s) = s + A_+(s) - A_-(s).$$

Method

- Plot $f(s)$ vs. s.
- Fixed points are intersections with the 45° line $y = s$.

Worked example (no shock, $\xi_t = 0$)

Let's test around two regions.

1. **Low** $s = 0$**:**
   $A_+(0) = 0,\ A_-(0) = 0 \Rightarrow f(0) = 0.$
   So $s^* = 0$ is a fixed point.

2. **Mid** $s = 4$**:**
   $A_+(4) = 0.9 \cdot 4/(1 + 0.8) = 3.6/1.8 = 2.0.$
   $A_-(4) = 0.45 \cdot 4 = 1.8.$
   So $f(4) = 4 + 2.0 - 1.8 = 4.2.$
   Slightly above the line → candidate stable point nearby.

3. **High** $s = 8$**:**
   $A_+(8) = 7.2/(1 + 1.6) = 7.2/2.6 \approx 2.77.$
   $A_-(8) = 0.15 \cdot 8 + 1.5 = 1.2 + 1.5 = 2.7.$
   So $f(8) = 8 + 2.77 - 2.7 \approx 8.07.$
   Very close to the line → another fixed point nearby.

Thus the map crosses the 45° line **three times**:

- A low point ($s \approx 0$),
- A mid unstable point,
- A high point ($s \approx 8$).

This is the classic "two basins separated by an unstable boundary."

5.4 Stability Test

Criterion: A fixed point s^* is **stable** if $|f'(s^*)| < 1$.

- Differentiate:

$$f'(s) = 1 + A'_+(s) - A'_-(s).$$

- Compute slopes piecewise.

Example calculations

- $A'_+(s) = \frac{0.9(1+0.2s) - 0.9 \cdot 0.2s}{(1+0.2s)^2} = \frac{0.9}{(1+0.2s)^2}$.
- For $s \leq 6$: $A'_-(s) = 0.45$.
- For $s > 6$: $A'_-(s) = 0.15$.

Now check:

1. **Low fixed point (near 0):**
   $A'_+(0) = 0.9/(1)^2 = 0.9$, $A'_- = 0.45$.
   So $f'(0) = 1 + 0.9 - 0.45 = 1.45$.
   Since $|1.45| > 1$, **unstable**.
2. **Middle fixed point (~4–5):**
   Say $s = 4.5$. $A'_+(4.5) = 0.9/(1 + 0.9)^2 = 0.9/(1.9)^2 \approx 0.25$.
   So $f'(4.5) = 1 + 0.25 - 0.45 = 0.80$.
   $|0.80| < 1 \rightarrow$ **stable**.
3. **High fixed point (~8):**
   $A'_+(8) = 0.9/(1 + 1.6)^2 = 0.9/2.6^2 = 0.9/6.76 \approx 0.13$.
   $A'_- = 0.15$.
   So $f'(8) = 1 + 0.13 - 0.15 = 0.98$.
   Borderline but $|0.98| < 1 \rightarrow$ **stable** (slow convergence).

5.5 Graphical Picture

How to read the graph (paper recommends drawing $f(s)$ vs. line $y = s$):

1. Horizontal axis = current pressure s.
2. Vertical axis = next period $f(s)$.
3. Intersections with diagonal $y = s$ = fixed points.
4. Steeper-than-45° (slope >1) → unstable; shallower → stable.

Interpretation.

- **Low regime:** resilient; shocks dampen out.
- **High regime:** hot throughput; fragile but self-sustaining until capacity kink bites.
- **Middle separator:** unstable; crossing it flips the basin of attraction.

5.6 Hysteresis and Two Basins

- **Hysteresis.** Once pushed into the "hot" basin, the system does not naturally return to the resilient one, unless a shock large enough drives it back below the unstable separator.
- **Plain words.** The system remembers which side of the unstable boundary it fell into.

This explains why crises and recoveries are asymmetric: **falling in is easier than climbing back out.**

Key Ideas (Section 5)

- A weighted scalar order parameter s_t captures system-wide stress.
- Simple S-shaped amplification + kinked absorption produce **three intersections**, i.e., **two stable regimes** plus one unstable separator.
- Stability test: check slope relative to 1 at fixed points.
- Hysteresis arises naturally from multiple basins.

Common Pitfalls

- Confusing "collapse to scalar" (compression) with crisis collapse (failure).
- Forgetting to test slope: not all fixed points are stable.
- Assuming shocks always decay: in the hot regime, they may persist or grow.

---

3 Quick Checks

1. **What makes a fixed point stable?**
   $|f'(s^*)| < 1$.
2. **How many long-run regimes exist in the toy map?**
   Two stable (resilient low, hot high), separated by one unstable.
3. **Why does hysteresis occur?**
   Because crossing the unstable separator changes the basin; returning requires a stronger counter-shock.

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6. Early-Warning Rules ("Black-Hole" Condition)

This section gives two **operational** flags—an AND rule with dwell time and a composite-score rule—to detect entry into the fragile, collapse-prone region. Both follow the paper's proposal to combine **CRP** (readiness), **SSE** (fragmentation), and optionally **max SSI** (local tension).

Symbols Box (first use in this section)

- $\theta_{crp}, \theta_{sse}, \theta_{ssi} \in (0,1)$: thresholds for CRP, SSE, and (optionally) max SSI.
- $D \in \mathbb{N}$: **dwell time** (consecutive periods the condition must hold).
- Ξ_t: composite early-warning score.
- $a, b, c > 0$: nonnegative weights for Ξ_t.
  All three indicators are defined in §2.

6.1 AND Rule with Dwell Time (BH-1)

Definition. Flag **BH** at time t if, **for D consecutive periods**,

$$\boxed{\mathrm{CRP}_\tau \geq \theta_{crp} \ \text{ AND } \ \mathrm{SSE}_\tau \geq \theta_{sse} \ \ (\text{AND optionally } \max_{i,k} \mathrm{SSI}_{i,\tau}^{(k)} \geq \theta_{ssi}) \ \text{ for all } \tau \in [t-D+1, t].}$$

Plain words. Risk is elevated **only when** (i) the system is primed to tip (CRP high), **and** (ii) narratives are fragmented (SSE high), **and** optionally (iii) at least one domain is locally tight (max SSI high), **persistently** for D steps.

Choosing thresholds and D.

- Practical choice is to set θ's near the **80–90th percentile** of each indicator's in-sample distribution and pick D to match your decision cycle (e.g., 3–4 weeks or 2 quarters). *(Contextual note consistent with the paper's falsifiable "flag must predict tails" guidance.)*

6.2 Composite Score Version (BH-2)

Definition. Form a weighted score

$$\boxed{\Xi_t = a \cdot \mathrm{CRP}_t + b \cdot \mathrm{SSE}_t + c \cdot \max_{i,k} \mathrm{SSI}_{i,t}^{(k)}.}$$

Flag **BH** if $\Xi_\tau \geq \theta$ for D **consecutive periods**.
Plain words. Combine the three into one dial and require it to stay in the red zone long enough to matter.

Weights. Start with $a = b = c = 1$ (equal importance) or set c higher if local overloads historically precede trouble in your domain. Calibrate θ so that historical BH time is rare (e.g., 5–10% of periods), then **test out-of-sample**. *(Follows the paper's "falsifiable mapping" insistence.)*

6.3 Numerical "Traffic-Light" Example

Assume weekly data, $D = 3$. Use $\theta_{crp} = 0.7$, $\theta_{sse} = 0.75$, $\theta_{ssi} = 0.8$. For the composite, use $a = b = c = 1$ and $\theta = 2.2$.

Week	CRP	SSE	max SSI	AND (BH-1) hold?	$\Xi = a\,\mathrm{CRP} + b\,\mathrm{SSE} + c\,\max \mathrm{SSI}$	Composite (BH-2) hold?
1	0.72	0.78	0.77	**No** (SSI < 0.80)	2.27	**Yes** ($\Xi \geq 2.2$)
2	0.74	0.80	0.82	**Yes**	2.36	**Yes**
3	0.69	0.82	0.84	**No** (CRP < 0.70)	2.35	**Yes**
4	0.71	0.76	0.83	**Yes**	2.30	**Yes**

Reading the table.

- **BH-1 (AND)** needs **three consecutive** weeks all true. Weeks 2 & 4 pass, but Week 3 fails CRP, so BH-1 **does not** trigger by Week 4.
- **BH-2 (composite)** exceeds θ from Week 1 through Week 4. With D=3, **BH-2 triggers at Week 3** and remains on at Week 4.

Interpretation.

- **AND** rule is stricter and reduces false alarms but may miss early signals.
- **Composite** is more sensitive; good for supervision with human review.

6.4 Validation & Use

- **Falsifiability test (as in the paper).** Periods with persistent BH flags must predict **higher tail-risk probabilities** (e.g., large output falls, defaults, layoffs) in a fixed forward window; otherwise the mapping is rejected or re-tuned.
- **Operationalization.** Monitor BH-1 for **hard alerts** (red light) and BH-2 for **advisories** (amber). Pair with policy playbooks (raise buffers, expand capacity, dampen risky conversions).

Key Ideas (Section 6)

- Combine **system readiness** (CRP), **narrative fragmentation** (SSE), and **local tension** (max SSI) to flag fragile regimes.
- Require **dwell time** so fleeting spikes don't trigger alarms.
- **Falsify** the mapping against tail events; retune thresholds/weights if it fails.

Common Pitfalls

- Picking thresholds from intuition only; use percentiles and **backtests**.
- Ignoring dwell time; single-week spikes are noisy.
- Treating BH as a forecast of timing; it's a **state flag** of fragility, not a clock.

---

3 Quick Checks

1. **What indicators does BH-1 combine?**
   CRP and SSE (optionally max SSI) with a dwell-time requirement.
2. **How do you set θ and D?**
   Use in-sample percentiles (e.g., 80–90th) and a dwell time matching decision cadence; then validate out-of-sample.
3. **What falsifies the BH design?**
   If persistent BH flags **do not** raise tail-risk probabilities at fixed horizons, the mapping is wrong and must be revised.

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7. Worked Examples (End-to-End Cases)

We now put the ingredients together in small, transparent cases. Each case computes **SSI**, **CRP**, **SSE**, and applies the **BH rules** with a short timeline. All constructions follow the uploaded paper's definitions and early-warning logic.

Symbols Box (for this section)

- $S_t^{(k)}$: surplus pressure in domain k at time t.
- $C_t^{(k)}$: absorption capacity in domain k.
- $\mathrm{SSI}_t^{(k)} = \min\left\{1, \frac{S_t^{(k)}}{S_t^{(k)} + C_t^{(k)} + \varepsilon}\right\}$.
- $\mathrm{CRP}_t = \sigma(w_\ell L_t + w_g G_t - w_b B_t + w_0)$.
- $\mathrm{SSE}_t = \frac{-\sum_j p_{t,j} \log p_{t,j}}{\log K}$.

- BH-1 (AND): CRP ≥ θ_{crp} **and** SSE ≥ θ_{sse} (**and** optionally max SSI ≥ θ_{ssi}) for D periods.
- BH-2 (composite): $\Xi_t = a\,\mathrm{CRP}_t + b\,\mathrm{SSE}_t + c\max \mathrm{SSI}_t \geq \theta$ for D periods.

Unless stated, use $\varepsilon = 0.1$. *"Contextual note" marks explanatory choices that are standard but not explicitly parameterized in the paper.*

Case A — Banking Stress (Leverage run-up with thinning buffers)

Setup (weekly): Track **Financial** F and **Material** M domains. Choose BH thresholds $\theta_{crp} = 0.70,\ \theta_{sse} = 0.75,\ \theta_{ssi} = 0.80,\ D = 3$. Composite $\theta = 2.2,\ a = b = c = 1$. *(Contextual note: thresholding by 80–90th percentiles is recommended as a practical rule-of-thumb.)*

Data (Weeks 1–4):

- Pressures/capacities → **SSI**
  - Week 1: $S_F = 50,\ C_F = 60 \Rightarrow \mathrm{SSI}_F = 50/(50 + 60 + 0.1) \approx 0.454$.
    $S_M = 20,\ C_M = 40 \Rightarrow \mathrm{SSI}_M \approx 0.333$.
  - Week 2: $S_F = 70,\ C_F = 60 \Rightarrow \mathrm{SSI}_F \approx 0.538$.
  - Week 3: $S_F = 90,\ C_F = 55 \Rightarrow \mathrm{SSI}_F \approx 0.620$.
  - Week 4: $S_F = 110,\ C_F = 50 \Rightarrow \mathrm{SSI}_F \approx 0.688$.
    $\max \mathrm{SSI} = \mathrm{SSI}_F$ each week.
- **CRP inputs** (standardized):
  L_t (leverage z-score) = (0.5, 0.9, 1.2, 1.4);
  G_t (valuation momentum) = (0.4, 0.6, 0.7, 0.8);
  B_t (buffers) = (1.0, 0.8, 0.6, 0.5).
  Weights: $(w_\ell, w_g, w_b, w_0) = (1.2, 0.8, 1.0, -0.2)$.
  → Scores:
  W1: $1.2(0.5) + 0.8(0.4) - 1.0(1.0) - 0.2 = 0.6 + 0.32 - 1.0 - 0.2 = -0.28 \Rightarrow \mathrm{CRP} \approx 0.430$.
  W2: $1.08 + 0.48 - 0.8 - 0.2 = 0.56 \Rightarrow 0.636$.
  W3: $1.44 + 0.56 - 0.6 - 0.2 = 1.20 \Rightarrow 0.768$.
  W4: $1.68 + 0.64 - 0.5 - 0.2 = 1.62 \Rightarrow 0.835$.
- **SSE** from topic shares (K=4).
  W1: $p = (0.60, 0.20, 0.10, 0.10) \Rightarrow \mathrm{SSE} \approx 0.64$.
  W2: $p = (0.45, 0.35, 0.10, 0.10) \Rightarrow \mathrm{SSE} \approx 0.86$.
  W3: $p = (0.40, 0.30, 0.20, 0.10) \Rightarrow \mathrm{SSE} \approx 0.93$.
  W4: $p = (0.35, 0.30, 0.20, 0.15) \Rightarrow \mathrm{SSE} \approx 0.96$. *(Contextual note: SSE rises as narratives fragment.)*

BH flags.

Week	CRP	SSE	max SSI	AND (BH-1)	Ξ	Composite (BH-2)
1	0.430	0.64	0.45	No	1.52	No
2	0.636	0.86	0.54	No	2.04	No
3	0.768	0.93	0.62	**CRP & SSE yes; SSI < 0.80**	2.32	**Yes (1/3)**
4	0.835	0.96	0.69	**CRP & SSE yes; SSI < 0.80**	2.49	**Yes (2/3 → triggers with D=2, not 3)**

Interpretation.
CRP and SSE cross their thresholds by Week 3, but local tightness (max SSI) does not, so **BH-1 stays off**. The composite passes θ from Week 3; with $D = 3$ it would trigger at Week 5 if conditions persist. Policy response (expand buffers, cap risky conversion M→F) targets CRP and Λ.

Case B — Supply Chain Congestion (S1 trigger, S3 mode switch)

Setup (monthly): One warehouse node (Material domain). Trigger at $\tau = 0.6$; mode switch at $\tau_m = 0.7$.

Month 1.

- Backlog $S^{(M)} = 120$, ship capacity $C^{(M)} = 180 \Rightarrow \text{SSI} = 120/(120 + 180 + 0.1) \approx 0.40 \rightarrow$ below trigger.
- Update (minimal law, no conversion): $S_2 = S_1 + \Gamma - A + 0 - L$.
  Let $\Gamma = 40$ (arrival), $A = \min\{S, C\} = 120$, $L = 0$. $\Rightarrow S_2 = 40$.

Month 2.

- Demand shock; arrivals jump: $\Gamma = 120$. Capacity falls (labor shortage) $C = 90$. Current $S = 40 \Rightarrow \text{SSI} = 40/(40 + 90 + 0.1) \approx 0.307$.
- But at end-of-month S surges to $S_3 = 40 + 120 - \min\{40, 90\} = 120$. At this new level, $\text{SSI} \approx 120/(120 + 90 + 0.1) \approx 0.571$.

Month 3.

- Another shock, $\Gamma = 150$, $C = 100$. Start $S = 120 \Rightarrow \text{SSI} \approx 0.545$ (still sub-trigger). After update, $S_4 = 120 + 150 - \min\{120, 100\} = 150 \Rightarrow \text{SSI} \approx 150/(150 + 100 + 0.1) \approx 0.600$.
- **S1 fires (hard)** next step (since SSI ≥ 0.6): add $\alpha = 30$ to Γ going forward.

Month 4 (overload mode).

- Because max SSI will exceed 0.7 during the month, switch to **overload** parameters (S3):
  - $\Gamma^{(\text{overload})} = \Gamma - 30$ (throttling),
  - $A^{(\text{overload})} = \min\{S, C^{\uparrow}\}$ with emergency shifts adding +40 to capacity,
  - $L^{(\text{overload})} = 20$ (markdowns/clearance).
- Start $S = 150$, baseline $\Gamma = 150 \rightarrow$ effective $\Gamma = 120$, $C = 100 + 40 = 140$.
  Update: $S_5 = 150 + 120 - \min\{150, 140\} + 0 - 20 = 150 + 120 - 140 - 20 = 110$.
  $\text{SSI}_5 \approx 110/(110 + 140 + 0.1) \approx 0.44$ (back under trigger).

Interpretation.
S1 ignites congestion once SSI crosses 0.6; S3's mode switch (throttling, emergency capacity, clearance) pulls the system back. This mirrors the paper's threshold + piecewise-capacity logic.

Case C — Platform Retention Shift (ads → quality reweighting)

Setup (weekly): Domains **Attention (A)** and **Cognitive (C)**. Overload in A triggers a policy shift (S3) that reduces ad generation and boosts absorption via UI changes.

Week 1.

- $S^{(A)} = 0.7$ hr, $C^{(A)} = 0.9 \Rightarrow \text{SSI}^{(A)} \approx 0.437$.
- $S^{(C)} = 0.6$ hr backlog, $C^{(C)} = 0.8 \Rightarrow \text{SSI}^{(C)} \approx 0.429$.
- $\Lambda_{C \to A} = 0.10$ (routing some decision friction into attention), loss $\delta = 0.2$. Small: $(1 - 0.2) \cdot 0.10 \cdot 0.6 = 0.048$ hr added to A.

Week 2 (marketing push).

- Ad generation $\Gamma^{(A)} = 0.5$. Absorption $A^{(A)} = \min\{S^{(A)}, C^{(A)}\} = 0.7$.

- Update $S^{(A)} \to 0.7 + 0.5 - 0.7 + 0.048 \approx 0.548$ hr.
- New $\mathrm{SSI}^{(A)} \approx 0.548/(0.548 + 0.9 + 0.1) \approx 0.36$ (still fine).

Week 3 (overstuffing).

- Extra campaign: $\Gamma^{(A)} = 0.9$. Start $S^{(A)} = 0.548$, $C^{(A)} = 0.9 \Rightarrow A^{(A)} = 0.548$.
- Update $S^{(A)} \to 0.548 + 0.9 - 0.548 + 0.048 = 0.948$ hr $\to \mathrm{SSI}^{(A)} \approx 0.948/(0.948 + 0.9 + 0.1) \approx 0.49$.

Week 4 (trigger and S3 mode switch).

- Suppose the platform's policy threshold is $\tau_m = 0.5$. With $\mathrm{SSI}^{(A)} \approx 0.49$ poised to exceed 0.5 mid-week due to organic growth, it flips to **overload** mode:
  - $\Gamma^{(A)} \downarrow$ (reduce ads; +quality weighting),
  - $A^{(A)} \uparrow$ (UI simplification increases effective capacity),
  - $\Lambda_{C \to A} \downarrow$ (less cognitive spillover).
- Concretely, set $\Gamma^{(A)} = 0.3,\ A^{(A)} = \min\{S,\ C^{(A)} + 0.4\} \Rightarrow C^{(A)} = 1.3$, and $\Lambda_{C \to A} = 0.04$.
- Update from $S^{(A)} = 0.948$:
  $S \to 0.948 + 0.3 - 0.948 + (1 - 0.2) \cdot 0.04 \cdot 0.6 \approx 0.948 + 0.3 - 0.948 + 0.019 \approx 0.319$ hr.
- $\mathrm{SSI}^{(A)} \approx 0.319/(0.319 + 1.3 + 0.1) \approx 0.19$ (healthy).

Interpretation.
A mode switch that **reweights from ads to quality** reduces generation, increases absorption, and weakens conversion from cognitive load. This matches the paper's S3 "piecewise parameter" design.

Case D — Limiting Case: Single Monotone Path (no competing basins)

Instruction to construct the limit (as in the paper):

> **Turn off type-conversion** ($\Lambda = 0$)**, kill buffers** ($C = 0$)**, and force reinvestment to be a monotone function of material surplus.** Then the reduced map is **monotone with one critical boundary**, i.e., **no competing basins**.

Concrete minimalist build.

- Single domain M (material).
- **No conversion:** $\Lambda = 0$.
- **No buffers/capacity:** $C = 0 \Rightarrow A(S; C) \equiv 0$.
- **Monotone reinvestment law:** $\Gamma(S) = g\,S$ with $g \in (0, 1)$ (fraction of surplus plowed back).
- **Losses:** constant fraction $d\,S$ with $d \in (0, 1)$.
- **Map:** $S_{t+1} = S_t + gS_t - 0 + 0 - dS_t = (1 + g - d)S_t$.

Behavior.

- If $1 + g - d < 1 \Rightarrow g < d$: S_t decays to 0.
- If $1 + g - d > 1 \Rightarrow g > d$: S_t grows without bound (until an external ceiling appears).
- **Critical boundary** at $g = d$: a single threshold splits decay vs growth.

- **Key point:** The function is **monotone** and there is **only one separator**—no middle unstable fixed point creating two basins. This realizes the paper's claim that the "single-path" view is a **special limiting region** of the general model.

*(Contextual note: you can add a soft ceiling—e.g., make losses increase with S—to create a single stable fixed point instead of divergence; it remains a **single-basin** picture without conversion and buffers.)*

Key Ideas (Section 7)

- End-to-end calculations use only: **generation − absorption + conversion − losses**, domain by domain.
- **Banking stress:** high CRP + rising SSE can flag fragility even when SSI is below a hard local threshold.
- **Supply chains/platforms:** S1 triggers and S3 mode switches operationalize congestion control and policy pivots.
- **Limiting case:** with $\Lambda = 0$, $C = 0$, and monotone reinvestment, dynamics reduce to a **single boundary**—no multi-basin behavior.

Common Pitfalls

- Ignoring **dwell time** in BH rules: single-period spikes are noisy and should not drive decisions.
- Treating conversion as free: routing can **relieve one domain but overheat another**.
- Forgetting that mode switches may alter **multiple** parameters at once, not just capacity.

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3 Quick Checks

1. **Which parameters do you change in an overload mode for a warehouse?**
   Decrease $\Gamma^{(M)}$ (throttle inflow), increase effective $C^{(M)}$ (emergency shifts), increase $L^{(M)}$ (clearance).
2. **What distinguishes the limiting single-path model from the general multi-basin model?**
   No conversion ($\Lambda = 0$), no buffers ($C = 0$), and monotone reinvestment remove the nonlinearity sources that create competing basins.
3. **Why can composite BH trigger sooner than the AND rule?**
   Because it aggregates signals; it can exceed θ even when one component (e.g., max SSI) is still below its individual threshold.

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8. Practice Set (with Answers)

Below are 12 undergrad-level exercises that practice the paper's core tools: **SSI, CRP, SSE, the minimal dynamic law, S1–S4, fixed points & stability, BH rules, and a simple S3 design.** All items are consistent with the uploaded paper's definitions.

Problems

1) Compute SSI (single domain)

A warehouse has backlog $S = 140$ pallets, ship capacity $C = 210$ pallets/day, $\varepsilon = 1$.
a) Compute SSI.
b) Interpret in one sentence.

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2) SSI across domains (max-SSI)

For a platform at week t:

- Attention: $S^{(A)} = 0.8$ hr, $C^{(A)} = 0.6$ hr, $\varepsilon = 0.02$.
- Cognitive: $S^{(C)} = 0.5$ hr, $C^{(C)} = 0.9$ hr, $\varepsilon = 0.02$.
  Compute both SSIs and $\max \mathrm{SSI}_t$.

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3) S1 trigger: hard vs soft

Let $\tau = 0.6,\ \alpha = 1.2$. For $\mathrm{SSI} = 0.58, 0.60, 0.70$:
a) Hard trigger increment ΔS.
b) Soft overshoot: $\Delta S = \alpha[\mathrm{SSI} - \tau]_+$.
c) Soft logistic: $\Delta S = \alpha\,\sigma(\beta(\mathrm{SSI} - \tau))$ with $\beta = 18$. Compare magnitudes.

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4) CRP calculation

Weights $(w_\ell, w_g, w_b, w_0) = (1.1, 0.7, 1.0, -0.1)$. Standardized inputs (L, G, B) for three weeks:
W1: $(0.3, 0.4, 1.1)$, W2: $(0.9, 0.6, 0.7)$, W3: $(1.2, 0.9, 0.6)$.
Compute CRP each week (logistic map $\sigma(x) = 1/(1 + e^{-x})$). Briefly interpret the trend.

---

5) SSE & effective topics

With $K = 5$ topic shares $p = (0.55, 0.20, 0.10, 0.10, 0.05)$:
a) Compute $H = -\sum p \log p$ and $\mathrm{SSE} = H/\log 5$.
b) Compute $N_{\mathrm{eff}} = e^H$. Explain what N_{eff} means.

---

6) Minimal dynamic law (two domains)

At time t: $S_t^{(M)} = 12,\ S_t^{(F)} = 5$.
Generation $\Gamma = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$; absorption $A = \begin{bmatrix} \min\{S^{(M)}, C^{(M)} = 10\} \\ \min\{S^{(F)}, C^{(F)} = 4\} \end{bmatrix}$; losses $L = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$.
Conversion matrix $\Lambda = \begin{bmatrix} 0.05 & 0.10 \\ 0.15 & 0.05 \end{bmatrix}$.
Compute S_{t+1}. Then check whether total pressure increased or decreased.

---

7) S4 routing with leakage

Start with $S^{(M)} = 20,\ S^{(F)} = 8$. Route $\lambda_{M \to F} = 0.25$ with loss $\delta_{M \to F} = 0.2$. Also route $\lambda_{F \to M} = 0.10$ with loss $\delta_{F \to M} = 0.5$.
How much is added to each destination and what fraction is lost in transit? Verify that routing alone cannot raise the **sum** $S^{(M)} + S^{(F)}$.

---

8) S3 mode switch table (design task)

Propose a **two-mode** parameter table for a call center (domains A & C). Fill:

- Normal: $\Gamma^{(A)}, A^{(A)}, \Lambda_{C \to A}, L^{(A)}$.
- Overload: (same symbols) but altered to relieve A-pressure.
  Use one sentence per parameter to justify the change (e.g., "UI simplification raises effective capacity").

---

9) Toy map: evaluate $f(s)$

Using the reduced map

$$f(s) = s + \frac{0.9s}{1 + 0.2s} - A_{-}(s), \quad A_{-}(s) = \begin{cases} 0.45s, & s \le 6 \\ 0.15s + 1.5, & s > 6 \end{cases}$$

compute $f(s)$ at $s = 0,\ 3,\ 6,\ 8$. For each, state whether $f(s)$ is above/below the 45° line.

---

10) Stability via slope

For the same map, compute $f'(s) = 1 + \frac{0.9}{(1 + 0.2s)^2} - A'_{-}(s)$ at $s = 0,\ 4.5,\ 8$ (with $A'_{-}(s) = 0.45$ for $s \le 6$ and 0.15 for $s > 6$).
Classify stability using $|f'(s^{*})| < 1$ as if each point were a fixed point.

---

11) BH rules on a 4-row panel

Thresholds: $\theta_{crp} = 0.70,\ \theta_{sse} = 0.80,\ \theta_{ssi} = 0.75,\ D = 3$. Composite: $a = b = c = 1,\ \theta = 2.20$.
Table:

t	CRP	SSE	max SSI
1	0.66	0.83	0.73
2	0.72	0.81	0.76
3	0.71	0.85	0.80
4	0.68	0.86	0.77

a) Does BH-1 (AND) trigger by $t = 4$?
b) Compute Ξ_t and decide if BH-2 triggers by $t = 4$ with $D = 3$.

---

12) Limiting case (single path, no competing basins)

Consider one domain with Λ=0, C=0, reinvestment $\Gamma(S) = gS$, losses dS.
a) Derive $S_{t+1} = (1 + g - d)S_t$.
b) Identify the single critical boundary separating decay vs growth.
c) Explain in one sentence why there is **no** multi-basin behavior in this limit.

Answer Key

1)

a) $\text{SSI} = \frac{140}{140 + 210 + 1} = \frac{140}{351} \approx 0.399$.
b) **Moderate tension**; capacity still exceeds pressure.

---

2)

Attention: $0.8/(0.8 + 0.6 + 0.02) = 0.8/1.42 \approx 0.563$.
Cognitive: $0.5/(0.5 + 0.9 + 0.02) = 0.5/1.42 \approx 0.352$.
max SSI $= 0.563$. (Attention is tighter.)

---

3)

a) Hard: SSI $= 0.58 \Rightarrow 0$; $0.60 \Rightarrow 1.2$; $0.70 \Rightarrow 1.2$.
b) Overshoot: $0,\ 1.2(0.00) = 0,\ 1.2(0.10) = 0.12$.
c) Logistic ($\beta = 18$):

- 0.58: $\sigma(18 \cdot -0.02) = \sigma(-0.36) \approx 0.410 \Rightarrow 0.492$.
- 0.60: $\sigma(0) = 0.5 \Rightarrow 0.6$.
- 0.70: $\sigma(1.8) \approx 0.858 \Rightarrow 1.029$.
  Order: hard ≥ logistic ≥ overshoot at/above threshold; below threshold, only logistic yields a partial bump.

---

4)

Linear scores:

- W1: $1.1(0.3) + 0.7(0.4) - 1.0(1.1) - 0.1 = 0.33 + 0.28 - 1.1 - 0.1 = -0.59 \Rightarrow \text{CRP} \approx 0.357$.
- W2: $1.1(0.9) + 0.7(0.6) - 1.0(0.7) - 0.1 = 0.99 + 0.42 - 0.7 - 0.1 = 0.61 \Rightarrow 0.648$.
- W3: $1.1(1.2) + 0.7(0.9) - 1.0(0.6) - 0.1 = 1.32 + 0.63 - 0.6 - 0.1 = 1.25 \Rightarrow 0.777$.
  Trend: readiness to tip rises across weeks (buffers fall; leverage/gains rise).

---

5)

a) $H = -(.55\ln.55 + .20\ln.20 + .10\ln.10 + .10\ln.10 + .05\ln.05) \approx 1.12$.
$\log 5 \approx 1.609 \Rightarrow \text{SSE} \approx 1.12/1.609 \approx 0.70$.
b) $N_{\text{eff}} = e^{1.12} \approx 3.06$.
Meaning: about three equally important topics would create the same entropy.

---

6)

$$A = \begin{bmatrix} \min\{12, 10\} = 10 \\ \min\{5, 4\} = 4 \end{bmatrix}, \quad \Lambda S_t = \begin{bmatrix} 0.05(12) + 0.10(5) = 1.1 \\ 0.15(12) + 0.05(5) = 2.05 \end{bmatrix}.$$

Then

$$S_{t+1} = \begin{bmatrix} 12 \\ 5 \end{bmatrix} + \begin{bmatrix} 6 \\ 2 \end{bmatrix} - \begin{bmatrix} 10 \\ 4 \end{bmatrix} + \begin{bmatrix} 1.1 \\ 2.05 \end{bmatrix} - \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 8.1 \\ 4.55 \end{bmatrix}.$$

Totals: 17 → 12.65 (down). Routing moved stress but row-sub-stochastic Λ prevented net creation.

---

7)

To F: $0.25(1 - 0.2) \cdot 20 = 0.20 \cdot 20 = 4$ added; $0.25 \cdot 20 = 5$ taken from M, so 1 lost in transit.
To M: $0.10(1 - 0.5) \cdot 8 = 0.05 \cdot 8 = 0.4$ added; $0.10 \cdot 8 = 0.8$ taken from F, so 0.4 lost.
Losses total 1.4. Sum before $= 28$, after routing $= 28 - 1.4 = 26.6$ (cannot rise under routing with leakage).

---

8) (sample design)

Normal: $\Gamma^{(A)} =$ 0.4 hr (ads), $A^{(A)} = \min\{S,\ C^{(A)}\}$ with $C^{(A)} = 1.0$, $\Lambda_{C \to A} = 0.10$ (mild spillover), $L^{(A)} = 0.05$ (drop rate).
Overload: $\Gamma^{(A)} = 0.15$ (throttle ads), $C^{(A)} = 1.4$ (UI simplification & batching), $\Lambda_{C \to A} = 0.03$ (less spillover), $L^{(A)} = 0.12$ (aggressive deferral/archive).
Justification (1-liners): throttle generation, expand capacity, reduce spillover, and increase controlled loss/deferral to clear queues.

---

9)

- $s = 0$: $A_+ = 0$, $A_- = 0 \Rightarrow f(0) = 0$ (on the line).
- $s = 3$: $A_+ = 2.7/(1 + 0.6) = 2.7/1.6 = 1.6875$, $A_- = 0.45 \cdot 3 = 1.35 \Rightarrow f(3) = 3 + 1.6875 - 1.35 = 3.3375$ (above line).
- $s = 6$: $A_+ = 5.4/(1 + 1.2) = 5.4/2.2 = 2.4545$, $A_- = 0.45 \cdot 6 = 2.7 \Rightarrow f(6) = 5.7545$ (below line).
- $s = 8$: $A_+ = 7.2/2.6 = 2.7692$, $A_- = 0.15 \cdot 8 + 1.5 = 2.7 \Rightarrow f(8) = 8.0692$ (slightly above line). Intersections implied near 0, between 3–6, and near 8.

---

10)

$f'(s) = 1 + \dfrac{0.9}{(1 + 0.2s)^2} - A'_-(s)$.

- $s = 0$: $A'_- = 0.45$, term $= 0.9 \Rightarrow f'(0) = 1 + 0.9 - 0.45 = 1.45$ (unstable if a fixed point).
- $s = 4.5$: $A'_- = 0.45$, term $= 0.9/(1 + 0.9)^2 = 0.9/3.61 \approx 0.249 \Rightarrow f'(4.5) \approx 0.799$ (stable).
- $s = 8$: $A'_- = 0.15$, term $= 0.9/(1 + 1.6)^2 = 0.9/6.76 \approx 0.133 \Rightarrow f'(8) \approx 0.983$ (stable but slow).

---

11)

a) AND needs CRP ≥0.70, SSE ≥0.80, max SSI ≥0.75 for **3 consecutive** periods.

- $t = 1$: CRP fails → no streak.
- $t = 2$: all pass (start streak = 1).
- $t = 3$: all pass (streak = 2).
- $t = 4$: CRP fails (0.68<0.70) → streak broken.
  Answer: BH-1 does **not** trigger by $t = 4$.
  b) Ξ:
- $t = 1$: $0.66 + 0.83 + 0.73 = 2.22$ (≥2.20).
- $t = 2$: 2.29 (≥).
- $t = 3$: 2.36 (≥).
- $t = 4$: 2.31 (≥).
  With $D = 3$, BH-2 **triggers at** $t = 3$ and remains on at $t = 4$.

---

12)

a) $S_{t+1} = S_t + gS_t - 0 + 0 - dS_t = (1 + g - d)S_t$.
b) Boundary at $g = d$.
c) The map is monotone with a **single separator** (no conversion, no buffers, monotone reinvestment), so no interior unstable fixed point → **no competing basins**.

Key Ideas (Section 8)

- **SSI/CRP/SSE** form a compact triad: local tightness, readiness to tip, and narrative fragmentation.
- **Minimal dynamic law** is bookkeeping with meaning: generate – absorb + convert – lose.
- **S1–S4** (thresholds, S-shapes, mode switches, routing) are enough to produce multi-basin behavior and

BH flags.

Common Pitfalls

- Forgetting to compare S to C (SSI), not just reading large S as "bad."
- Building CRP from non-standardized inputs, making weights meaningless.
- Treating routing as harmless: it can **raise local** risk even if totals don't rise.

---

3 Quick Checks

1. Which indicator best captures system-level priming to tip? **CRP.**
2. What does max SSI add that CRP and SSE don't? **A direct view of the tightest local bottleneck.**
3. In the limiting single-path case, what removes multi-basin behavior? **No conversion, no buffers, monotone reinvestment.**

---

9. Implementation Appendix (Python Snippets)

Minimal NumPy/Matplotlib code to compute **SSI**, **CRP**, **SSE**, simulate the **toy map**, make the $f(s)$ **vs** $y = s$ plot, and apply **early-warning flags**. Functions mirror the definitions used throughout and match the uploaded paper.

> Install needs: only `numpy` and `matplotlib`. No web calls.
> All snippets are self-contained; copy–paste into one `.py` or a notebook cell.

9.1 Core Indicators: SSI, CRP, SSE

```
import numpy as np
import math

def ssi(S, C, eps=1e-3):
    """
    Surplus Saturation Index (SSI) for arrays or scalars.
    SSI = min(1, S / (S + C + eps)) in [0, 1].
    Inputs can be floats or NumPy arrays (broadcasted).
    """
    frac = np.divide(S, (S + C + eps))
    return np.minimum(1.0, np.maximum(0.0, frac))

def logistic(x):
    return 1.0 / (1.0 + np.exp(-x))

def crp(L, G, B, w_l=1.0, w_g=1.0, w_b=1.0, w0=0.0):
    """
    Collapse Readiness Potential (CRP):
    CRP = sigmoid(w_l*L + w_g*G - w_b*B + w0) in [0,1].
    L,G,B should be standardized or percentile-mapped first.
    """
    score = w_l*L + w_g*G - w_b*B + w0
    return logistic(score)

def sse(p):
    """
    Semantic (normalized Shannon) Entropy:
    SSE = [-sum p_j log p_j] / log K in [0,1].
    p: iterable of shares summing to 1 (clips tiny vals for safety).
    Returns (SSE, H, K) where H is unnormalized entropy.
    """
    p = np.asarray(p, dtype=float)
```

```
    p = np.clip(p, 1e-12, 1.0)
    p = p / p.sum()
    H = -np.sum(p * np.log(p))
    K = len(p)
    SSE = H / np.log(K)
    return float(SSE), float(H), K
```

Plain-language check.

- `ssi`: compares pressure vs capacity in one fraction (near 1 = tight).
- `crp`: blends leverage & gains (↑) and buffers (↓) then squashes to [0,1].
- `sse`: measures topic fragmentation; near 1 = many equally sized narratives.

9.2 Toy Map: Definition, Simulation, Plot $f(s)$ vs $y = s$

```
import numpy as np
import matplotlib.pyplot as plt

def A_plus(s):
    """S-shaped amplification: A_+(s) = 0.9 s / (1 + 0.2 s)"""
    return 0.9*s / (1.0 + 0.2*s)

def A_minus(s):
    """Piecewise absorption:
       0.45 s,  s <= 6
       0.15 s + 1.5, s > 6
    """
    s = np.asarray(s)
    out = np.where(s <= 6.0, 0.45*s, 0.15*s + 1.5)
    return out

def f_map(s):
    """f(s) = s + A_+(s) - A_-(s)  (no shock term)"""
    return s + A_plus(s) - A_minus(s)

def simulate_toy_map(s0, T=50, xi_std=0.0, rng=None):
    """
    Simulate s_{t+1} = s_t + A_+(s_t) - A_-(s_t) + xi_t
    with optional Gaussian noise xi_t ~ N(0, xi_std^2).
    """
    if rng is None:
        rng = np.random.default_rng(0)
    s = np.zeros(T+1)
    s[0] = float(s0)
    for t in range(T):
        shock = rng.normal(0.0, xi_std) if xi_std > 0 else 0.0
        s[t+1] = s[t] + A_plus(s[t]) - A_minus(s[t]) + shock
        s[t+1] = max(0.0, s[t+1])  # keep nonnegative
    return s

def plot_f_vs_y(s_min=0.0, s_max=12.0, n=400):
    """
    Plot f(s) and the 45° line y=s.
    How to read:
      1) Intersections are fixed points.
      2) Where f(s) is above y=s, s_{t+1} > s_t (s drifts up).
      3) Slopes < 1 near intersections indicate stability.
    """
    s = np.linspace(s_min, s_max, n)
    f = f_map(s)
    plt.figure(figsize=(6,4))
    plt.plot(s, f, label="f(s)")
    plt.plot(s, s, linestyle="--", label="y = s")
    plt.xlabel("s (current)")
    plt.ylabel("f(s) (next)")
    plt.title("Toy Map: f(s) vs. 45° line")
    plt.legend()
    plt.tight_layout()
    plt.show()
```

How to read the graph (3 bullets).

1. Intersections of `f(s)` with `y=s` are fixed points.
2. If `f(s)` lies **above** the diagonal, the next value is larger → upward drift; **below** → downward drift.
3. A slope magnitude less than 1 at an intersection usually indicates stability.

Quick demo (optional in a notebook):

```
# Plot and simulate from two initials to see two basins
plot_f_vs_y()
s_low = simulate_toy_map(s0=0.5, T=30)
s_high = simulate_toy_map(s0=9.0, T=30)
print("Final s (low init):", s_low[-1])
print("Final s (high init):", s_high[-1])
```

9.3 Early-Warning Flags (BH-1 and BH-2)

```
def bh1_and_rule(CRP, SSE, max_SSI=None,
                 theta_crp=0.7, theta_sse=0.8, theta_ssi=0.8, D=3):
    """
    BH-1 (AND) with dwell time D.
    Requires CRP>=theta_crp AND SSE>=theta_sse,
    and optionally max_SSI>=theta_ssi if max_SSI is provided.
    Returns a Boolean array of same length as inputs.
    """
    CRP = np.asarray(CRP); SSE = np.asarray(SSE)
    assert CRP.shape == SSE.shape
    cond = (CRP >= theta_crp) & (SSE >= theta_sse)
    if max_SSI is not None:
        max_SSI = np.asarray(max_SSI)
        assert max_SSI.shape == CRP.shape
        cond = cond & (max_SSI >= theta_ssi)
    # dwell time: rolling all-true for last D periods
    out = np.zeros_like(cond, dtype=bool)
    streak = 0
    for t, ok in enumerate(cond):
        streak = streak + 1 if ok else 0
        out[t] = (streak >= D)
    return out

def bh2_composite(CRP, SSE, max_SSI, a=1.0, b=1.0, c=1.0, theta=2.2, D=3):
    """
    BH-2 composite score Xi = a*CRP + b*SSE + c*max_SSI.
    Flag if Xi>=theta for D consecutive periods.
    Returns (Xi array, Boolean flags).
    """
    CRP = np.asarray(CRP); SSE = np.asarray(SSE); max_SSI = np.asarray(max_SSI)
    Xi = a*CRP + b*SSE + c*max_SSI
    out = np.zeros_like(Xi, dtype=bool)
    streak = 0
    for t, val in enumerate(Xi):
        if val >= theta:
            streak += 1
        else:
            streak = 0
        out[t] = (streak >= D)
    return Xi, out
```

Plain-language check.

- `bh1_and_rule`: strict gate—needs **all** indicators above thresholds for **D** steps.
- `bh2_composite`: single dial Ξ; above a cutoff for **D** steps.

Tiny example:

```
CRP = [0.66, 0.74, 0.71, 0.68]
SSE = [0.83, 0.81, 0.85, 0.86]
mSSI= [0.73, 0.76, 0.80, 0.77]
```

```
bh1 = bh1_and_rule(CRP, SSE, mSSI, theta_crp=0.70, theta_sse=0.80, theta_ssi=0.75, D=3)
Xi, bh2 = bh2_composite(CRP, SSE, mSSI, a=1, b=1, c=1, theta=2.20, D=3)

print("BH-1 flags:", bh1.tolist())
print("Xi:", [round(x,2) for x in Xi], "BH-2 flags:", bh2.tolist())
```

Expected (matching Section 8's style): BH-1 likely **does not** trigger by the last row; BH-2 likely **does** once the 3-period dwell is met.

9.4 Optional helpers (conversion & minimal law bookkeeping)

```
def apply_conversion(S, Lambda, delta=None):
    """
    S: (K,) vector of domain pressures.
    Lambda: (K,K) routing fractions, row-sub-stochastic (rows sum <= 1).
    delta: (K,K) loss fraction per route (same shape), default 0.
    Returns new S after routing with leakage.
    """
    S = np.asarray(S, dtype=float)
    K = len(S)
    Lmb = np.asarray(Lambda, dtype=float)
    if delta is None:
        delta = np.zeros_like(Lmb)
    else:
        delta = np.asarray(delta, dtype=float)
    # outbound amounts by source j: Lmb[j,*] * S[j]
    routed = (Lmb * (1.0 - delta)) * S[:, None]
    incoming = routed.sum(axis=0)
    outbound = (Lmb * S[:, None]).sum(axis=1)
    leakage = (Lmb * delta * S[:, None]).sum()
    S_after = S - outbound + incoming
    return S_after, leakage

def minimal_update(S, Gamma, A, Lambda=None, L=None, delta=None):
    """
    One-step update: S_{t+1} = S + Gamma - A + Lambda*S - L,
    implemented as: S + Gamma - A - outbound + incoming - L.
    Gamma, A, L are (K,) arrays. Lambda is (K,K).
    """
    S = np.asarray(S, dtype=float)
    Gamma = np.asarray(Gamma, dtype=float)
    A = np.asarray(A, dtype=float)
    L = np.zeros_like(S) if L is None else np.asarray(L, dtype=float)

    if Lambda is None:
        S_conv = S.copy()
    else:
        S_conv, _ = apply_conversion(S, Lambda, delta=delta)

    return S + Gamma - A + (S_conv - S) - L
```

Plain-language check.

- `apply_conversion`: routes stress across domains with optional leakage; totals can only fall or stay the same (never rise from routing alone).
- `minimal_update`: exact bookkeeping of **generate − absorb + convert − lose**.

9.5 Sanity Tests (quick)

```
# SSI sanity
print("SSI(120,180) ≈", round(float(ssi(120,180,eps=1)), 3))  # ~0.399

# CRP sanity
print("CRP demo:", round(float(crp(1.0, 0.5, 0.8, 1.2, 0.8, 1.0, -0.2)), 3))  # ~0.645

# SSE sanity
SSE, H, K = sse([0.45,0.45,0.05,0.05])
```

```
print("SSE ~", round(SSE, 2), "Neff ~", round(math.e**H, 2))  # ~0.81, ~3.06

# Toy map fixed points (visual)
plot_f_vs_y()
```

---

10. Glossary & Symbol Index

A one-page-ish reference for every symbol used, with ranges, units, and plain-English meanings. All definitions follow the uploaded paper's notation and constructs.

Core State, Capacity, and Indices

Symbol	Range / Units	Meaning & How to Read
i	index	Unit (sector, firm, platform node).
$k \in \{\mathrm{M}, \mathrm{F}, \mathrm{I}, \mathrm{A}, \mathrm{C}\}$	set	Domains: **M**aterial, **F**inancial, **I**nstitutional, **A**ttention, **C**ognitive.
t	periods	Discrete time (week, month, quarter).
$S_{i,t}^{(k)} \geq 0$	domain units	**Surplus pressure** in domain k for unit i (e.g., inventories, leverage strain, rule overload, ad load, decision backlog). Bigger = tighter.
$C_{i,t}^{(k)} \geq 0$	same as S	**Absorption capacity** in domain k: how much stress the domain can absorb/ dissipate. Higher C = easier relief.
$\varepsilon > 0$	small, unitless	Numerical guard to avoid divide-by-zero in SSI. Typical $\varepsilon \in [10^{-3}, 10^{-1}]$.
$\mathrm{SSI}_{i,t}^{(k)} \in [0,1]$	unitless	**Surplus Saturation Index**: $\min\{1, \frac{S}{S+C+\varepsilon}\}$. Near 0 = slack; near 1 = at/over capacity.
$\mathrm{CRP}_t \in [0,1]$	unitless	**Collapse Readiness Potential**: logistic blend of leverage L_t and short-run gains G_t (↑ risk) minus buffers B_t (↓ risk). Higher = readier to tip.
$\mathrm{SSE}_t \in [0,1]$	unitless	**Semantic (normalized Shannon) Entropy** of topic shares $p_{t,j}$. Higher = more fragmented narratives (harder coordination).

Drivers, Flows, and Bookkeeping

Symbol	Range / Units	Meaning & How to Read
S_t	vector $\mathbb{R}_{\geq 0}^K$	Stacked domain pressures at t.
$\Gamma(X_t) \geq 0$	vector, like S	**Generation** of new pressure (production overshoot, valuation run-up, new rules, ad push, tasks arrival).
$A(S_t; C_t) \geq 0$	vector	**Absorption/dissipation** using capacities/buffers; often piecewise or saturating. Larger $C \Rightarrow$ typically larger A.
Λ	$K \times K$ matrix	**Conversion matrix** (row-sub-stochastic: row sums ≤ 1): routes stress across domains without creating net pressure.
$\delta_{j \to k} \in [0,1]$	fraction	**Conversion loss** for routing $j \to k$. Losses dissipate pressure in transit.
$L_t \geq 0$	vector like S	**Losses/write-offs** (liquidations, clearances, amnesties, task drops). Reduces pressure but may have costs.
Minimal law	—	$S_{t+1} = S_t + \Gamma(X_t) - A(S_t; C_t) + \Lambda S_t - L_t$. "Have + generate − absorb + convert − lose."

Indices: Ingredients and Parameters

Symbol	Range / Units	Meaning & How to Read
L_t	standardized	**Leverage metric** (e.g., debt/collateral, margin usage). Part of CRP.

Symbol	Range / Units	Meaning & How to Read
G_t	standardized	**Short-run amplification** (valuation/throughput momentum, self-reinforcing loop proxy). In CRP.
B_t	standardized	**Buffers** (capital, liquidity, inventory slack, throttling/automation). Enters CRP with a minus sign.
w_ℓ, w_g, w_b, w_0	real	CRP weights and intercept in $\sigma(w_\ell L + w_g G - w_b B + w_0)$. Calibrate from data or expert judgment.
$\sigma(x)$	$[0,1]$	Logistic function $1/(1+e^{-x})$. Used in CRP and soft triggers.
$p_{t,j}$	$[0,1]$, $\sum p = 1$	Topic-cluster shares for SSE.
K	$\mathbb{N}$	Number of topic clusters.
H_t	≥ 0	Shannon entropy $H = -\sum p \log p$ (unnormalized).
N_{eff}	≥ 1	Effective number of topics e^H. "As-if" count of equally-sized topics.

S-Mechanisms (Nonlinearities)

Symbol	Range / Units	Meaning & How to Read
$\alpha_{i,k} > 0$	increment	S1 **trigger size** added to pressure when local SSI crosses threshold.
$\tau_{i,k} \in (0,1)$	threshold	S1 **trigger threshold** on SSI. Hard: step; Soft: overshoot/logistic.
$A_+(s)$	≥ 0	S2 **amplification** (e.g., $rs/(1+\alpha s)$): small at low s, ramps up, saturates.
$m_t \in \{\text{normal}, \text{overload}\}$	mode	S3 **role/structure shift**: parameters (Γ, A, Λ, L) switch piecewise at a stress threshold.
$\lambda_{j\to k} \in [0,1]$	fraction	S4 **routing share** from j to k; row sums ≤ 1. With δ, allows leakage.

Scalar Reduction & Toy Map

Symbol	Range / Units	Meaning & How to Read
$w \geq 0$	weights	Domain weights used to compress S_t to a scalar.
$s_t = w^\top S_t \geq 0$	scalar	**Order parameter** (scalar compression of multi-domain pressure). "Collapse to scalar" = dimensionality reduction (not a crisis).
$A_+(s)$	≥ 0	In toy map: $0.9s/(1+0.2s)$. S-shaped gain.
$A_-(s)$	≥ 0	In toy map: $0.45s$ (if $s \leq 6$); $0.15s + 1.5$ (if $s > 6$). Kinked absorption.
ξ_t	small shock	Disturbance term in reduced map.
$f(s)$	—	$s + A_+(s) - A_-(s)$. Fixed points where $f(s) = s$. Stability by (

Early-Warning ("Black-Hole") Flags

Symbol	Range / Units	Meaning & How to Read
$\theta_{crp}, \theta_{sse}, \theta_{ssi} \in (0,1)$	thresholds	CRP/SSE/(optional) max-SSI cutoffs (often 80–90th percentiles).
$D \in \mathbb{N}$	periods	**Dwell time**: must hold for D consecutive periods to flag.
Ξ_t	real	Composite score $a\,\text{CRP} + b\,\text{SSE} + c\,\max\text{SSI}$. Flag if $\Xi \geq \theta$ for D periods.
a, b, c, θ	real	Weights & cutoff for composite rule. Calibrate, then falsify on tail-risk prediction.

Crosswalk: Classical "Surplus" Talk → This Framework's "Excess-Pressure" Language

The paper generalizes a traditional single-surplus lens into multi-domain **excess pressure** that is measured relative to **absorption capacity** and allowed to **convert** across domains. Below is a neutral terminology bridge, avoiding school-specific labels while preserving meaning.

Classical phrase (neutralized)	Excess-pressure counterpart here
"Surplus beyond immediate needs (goods or claims)"	**Surplus pressure** $S^{(k)}$ in the relevant domain (material, financial, institutional, attention, cognitive).
"Reinvestment / expansion of activity"	**Generation** $\Gamma(X_t)$ that adds new pressure (production, lending, rule creation, messaging, tasks).
"Capacity to absorb shocks / slack"	**Absorption capacity** $C^{(k)}$ and **absorption function** $A(S; C)$. Higher $C \Rightarrow$ more dissipation.
"Channels/circuits where pressures shift form"	**Conversion matrix** Λ (routing with losses δ); moves stress across domains but cannot create net pressure (rows ≤ 1).
"Liquidation / write-downs / abandonment"	**Losses** L_t that permanently remove pressure (possibly costly).
"Boom-bust self-reinforcement"	**Amplification** $A_+(s)$ (S-shaped) and **kinked absorption** $A_-(s) \rightarrow$ **multiple attractors** & hysteresis.
"System fragility / tipping readiness"	**CRP** (readiness), **SSE** (fragmentation), **max SSI** (local bottleneck) and **BH** dwell-time rules.
"A single necessary path (limit case)"	Turn off conversion (Λ=0), kill buffers (C=0), enforce monotone reinvestment → **one critical boundary**, no competing basins.

Key Ideas (Section 10)

- **Everything is pressure vs capacity.** SSI quantifies local tension; CRP and SSE capture system priming and coordination frictions.
- **Accounting, not alchemy.** Conversion respects row-sub-stochasticity; routing cannot create total pressure.
- **Simple shapes suffice.** Thresholds + S-shapes + piecewise kinks → multiple regimes and testable early-warnings.

Common Pitfalls

- Reading big S as danger **without** checking C (SSI matters).
- Building CRP from raw, incomparable units (standardize or use percentiles first).
- Forgetting dwell time: BH flags are **state** signals, not clocks.

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3 Quick Checks

1. **What keeps conversion from creating pressure out of thin air?**
   Row-sub-stochastic Λ (row sums ≤ 1) and explicit leakage δ.
2. **Which indicator best captures narrative fragmentation?**
   SSE (normalized entropy of topic shares).
3. **How do you read** SSI ≈ 1**?**
   Pressure is at/over capacity: triggers and spillovers are likely unless capacity, buffers, or losses change.

Cheat Sheet — *From Surplus Dynamics* (Undergrad Recap)

All definitions and formulas follow the uploaded paper.

Core Formulas (with 1-line readings)

SSI (local tension)

$$\mathrm{SSI}_{i,t}^{(k)} = \min\left\{1, \frac{S_{i,t}^{(k)}}{S_{i,t}^{(k)} + C_{i,t}^{(k)} + \varepsilon}\right\} \in [0,1].$$

Pressure vs capacity. Near 1 ⇒ tight/overloaded.

CRP (system readiness)

$$\mathrm{CRP}_t = \sigma\left(w_\ell L_t + w_g G_t - w_b B_t + w_0\right), \quad \sigma(x) = \frac{1}{1+e^{-x}}.$$

Leverage & short-run gains raise risk; buffers lower it; squashed to $[0,1]$.

SSE (narrative fragmentation)

$$\mathrm{SSE}_t = \frac{-\sum_{j=1}^{K} p_{t,j} \log p_{t,j}}{\log K} \in [0,1], \quad N_{\mathrm{eff}} = e^{H}.$$

Closer to 1 ⇒ more fragmented narratives, harder coordination.

Minimal dynamic law (bookkeeping)

$$S_{t+1} = S_t + \Gamma(X_t) - A(S_t; C_t) + \Lambda S_t - L_t.$$

Have + generate − absorb + convert − lose. Λ moves stress but cannot create it on net.

S1 Trigger (hard/soft)

$$\Delta S^{(k)} \mathrel{+}= \alpha_{i,k}\, \mathbf{1}\{\mathrm{SSI} \geq \tau\} \quad \text{or} \quad \alpha_{i,k}[\mathrm{SSI} - \tau]_+ \quad \text{or} \quad \alpha_{i,k}\, \sigma(\beta(\mathrm{SSI} - \tau)).$$

Turns on extra pressure when local tension crosses a threshold.

S2 Feedback (amplification, saturating)

$$A_+(s) = \frac{r\, s}{1 + \alpha s} \quad \text{(or logistic variant)}.$$

Self-reinforcement that ramps up then flattens.

S3 Role/Structure shift (mode switch)

$$(\Gamma, A, \Lambda, L) = \begin{cases} (\cdot)^{(\mathrm{normal})}, & \max \mathrm{SSI} < \tau_m \\ (\cdot)^{(\mathrm{overload})}, & \text{otherwise} \end{cases}$$

Parameters flip when stress is high (throttle, add buffers, raise losses, reroute).

S4 Conversion (routing with losses)

$$S^{(k)} \leftarrow S^{(k)} + \sum_j \lambda_{j\to k}(1 - \delta_{j\to k}) S^{(j)}, \quad \sum_k \lambda_{j\to k} \leq 1.$$

Routes stress across domains; leakage allowed; rows ≤ 1 prevent net creation.

One-dimensional toy map (two basins)

$$s_{t+1} = s_t + A_+(s_t) - A_-(s_t) + \xi_t, \quad A_+(s) = \frac{0.9s}{1 + 0.2s},\ A_-(s) = \begin{cases} 0.45s, & s \leq 6 \\ 0.15s + 1.5, & s > 6. \end{cases}$$

Plot $f(s)$ vs $y = s$; intersections are fixed points; $|f'(s^{\ast})| < 1 \Rightarrow$ stable.

Early-warning ("BH")

- **AND:** CRP $\geq \theta_{crp}$ **and** SSE $\geq \theta_{sse}$ (and optionally maxSSI $\geq \theta_{ssi}$) for D periods.
- **Composite:** $\Xi_t = a\,\mathrm{CRP} + b\,\mathrm{SSE} + c\,\max \mathrm{SSI} \geq \theta$ for D periods.
  Set cutoffs near 80–90th percentiles; validate on tail events.

Key Concepts (at a glance)

- **Excess pressure vs capacity:** Risk is about S relative to C, not the size of S alone. (SSI)
- **System readiness:** Leverage/amplification minus buffers ⇒ CRP. High CRP = primed to tip.
- **Narrative friction:** Fragmented discourse (SSE↑) weakens coordination/absorption.
- **Routing law:** Λ rows ≤ 1; conversion shifts where stress sits without raising the total. Leakage δ dissipates.
- **Multiple attractors:** S-shaped gains + kinked absorption ⇒ two stable regimes separated by an unstable boundary (hysteresis).
- **Two meanings of "collapse":** (i) **to scalar** $s_t = w^{\top} S_t$ = compression; (ii) **tipping** into the hot/fragile basin.
- **Limiting single-path case:** If Λ=0, C=0, and reinvestment is monotone in material surplus, the reduced map is monotone with **one** critical boundary—no competing basins.

Symbol Glossary (micro)

- **Indices:** i unit; $k \in \{M, F, I, A, C\}$ domain; t time.
- **States/Capacity:** $S_{i,t}^{(k)}$ pressure; $C_{i,t}^{(k)}$ absorption; ε small guard.
- **Indicators:** SSI local tension; CRP readiness; SSE fragmentation; $p_{t,j}$ topic share; K topics; $N_{\mathrm{eff}} = e^{H}$.
- **Drivers (CRP):** L_t leverage; G_t short-run gain; B_t buffers; w_{ℓ}, w_g, w_b, w_0 weights; σ logistic.
- **Flows:** Γ generation; $A(\cdot)$ absorption; Λ conversion (rows ≤ 1); δ conversion loss; L_t losses.
- **Nonlinearities:** α trigger size; τ, β trigger threshold/slope; $A_{+}(s), A_{-}(s)$ gain/absorption shapes; m_t mode.
- **Scalar reduction:** w weights; $s_t = w^{\top} S_t$; $f(s) = s + A_{+}(s) - A_{-}(s)$; stability $|f'(s^{\ast})| < 1$.
- **BH settings:** $\theta_{crp}, \theta_{sse}, \theta_{ssi}$ cutoffs; D dwell; $\Xi = a\,\mathrm{CRP} + b\,\mathrm{SSE} + c\,\max \mathrm{SSI}$; θ composite cutoff.

Done. Keep this alongside the Python appendix to compute the indicators, plot $f(s)$, and apply BH flags.

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appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.